

Impact Statement

Latent Probing for Mental Health Sentiment Classification

Purpose of the Project

Our project, Latent Probing for Mental Health Sentiment Classification, addresses a real gap in mental health support: early signs of psychological distress, expressed subtly in everyday language, often go unnoticed until a crisis emerges. This is especially true for people who avoid seeking help due to stigma or limited access to care. Using latent probing techniques on text, our model detects patterns associated with Depression, Anxiety, and Stress, aiming to support, not replace, early identification efforts within a community.

Positive Impacts

For individuals, the tool could help surface early warning signs in their own writing before a crisis develops, prompting earlier self reflection or outreach for support. For communities and organizations such as school counseling programs or peer support groups, it could act as a low cost screening aid to help direct limited human attention toward people who may need it most. More broadly, by keeping the focus on early, non-crisis distress rather than acute risk, the project supports a shift toward preventative rather than reactive mental health care.

Potential Risks and Harms

The most serious risk is over-reliance on the model's output as if it were a diagnosis, which it is not; a false positive or false negative could lead to inappropriate action or a missed opportunity for support. There is also a privacy risk, since mental health related text is highly sensitive, and a fairness risk, since the training data likely underrepresents non-English speakers, older adults, and people without regular social media access, meaning the tool may work less well for exactly the populations most affected by care gaps.

Values

The project is guided by privacy, safety, fairness, and transparency. Privacy is protected by using only secondary, anonymized public data rather than collecting new personal data. Safety is supported by keeping human judgment central to any decision made using the model's output. Fairness is pursued by openly documenting known representation gaps rather than hiding them. Transparency comes from the choice of latent probing itself, which produces interpretable confidence signals rather than an unexplained black box output.

Mitigation Strategies

The team will validate a sample of dataset labels against a DSM-5-based rubric, consult a psychology or counseling professional to review this rubric, clearly communicate that the tool is a support aid rather than a diagnostic tool, and restrict use to educational and research purposes rather than public deployment at this stage.

Limitations and Uncertainties

This project relies on an existing public dataset whose original contributors were not asked for consent for this specific research use, and whose demographic makeup is not fully known. As such, the model's

real-world performance across different populations, languages, and cultural contexts remains uncertain and would require further testing before any broader use.